

Chapter 123 — Surgical Procedures for Sphincteric Incontinence in the Male: The Artificial Urinary Sphincter and Perineal Sling Procedures

- **Male stress urinary incontinence (SUI)**, the involuntary leakage of urine, **substantially impairs quality of life (QoL)**. Causes: congenital anomalies, injury, genitourinary surgery, other conditions.
- **Radical prostatectomy (RP) for prostate cancer (CaP) is the leading cause.** Mechanism: **failure to store because external sphincter resistance is insufficient.**
- **The US burden of post-RP urinary incontinence (UI) remains high** because of RP volume, but is **expected to fall**. Reasons: better technique; fewer RPs after the **2012 US Preventive Services Task Force (USPSTF) recommendation against prostate-specific antigen (PSA) screening**; more **active surveillance**.
- **QoL impact drives treatment-seeking.** Perineal slings and the artificial urinary sphincter (AUS) work by **increasing outlet resistance.**
- **Barriers to care are physician and patient perceptions. Decisional regret is highest after conservative treatment**, which argues for outreach.
- **The AUS is the gold standard, effective for all degrees of SUI. Bulbar urethral slings**, now several generations old with long-term data, suit **mild-to-moderate UI.**

CLASSIFICATION, PATHOPHYSIOLOGY, AND ETIOLOGY

- **Continence needs the bladder, the internal sphincter (bladder neck and prostatic urethra) and the external sphincter (rhabdosphincter).** Male SUI (leakage with raised abdominal pressure) occurs **only when both sphincters are impaired.**

INTERNAL VERSUS EXTERNAL SPHINCTER INCOMPETENCE — CAUSES

Internal sphincter	External sphincter
Pelvic surgery or radiation	Radical prostatectomy (commonest)
Bladder neck injury	Pelvic radiation; pelvic fracture urethral injury
Specific sympathetic neuropathic dysfunction	Traumatic and acquired myelopathy
Embryological disruption	Spinal dysraphism, sacral agenesis, exstrophy–epispadias complex

- **Reduced compliance, detrusor overactivity (DO) and urgency urinary incontinence (UUI) complicate diagnosis** and demand careful evaluation.
- **Prevalence: about 11% of men aged 60–64 and 31% over 85.** Incidence tracks the causative procedures.
- **Historical post-RP UI ranges from 2.5% to 87%.** The spread reflects patient factors, surgeon factors and method. **Self-report gives higher rates than physician-recorded outcomes.**
- **Continence improves progressively for up to 2 years after RP.**
- **Nerve sparing reduces post-RP SUI**, probably by preserving sensory and motor pudendal innervation of the rhabdosphincter. It gives **faster return, not better final continence** (meta-analyses).
- **Autonomic afferent denervation and impaired membranous urethral sensitivity** are associated with SUI.
- **Other mechanisms:** damage to the rhabdosphincter, pelvic floor or bladder neck. Incontinent men differ in **urethral pressure profile, length and sphincter morphology.**
- **Posterior reconstruction speeds recovery but does not improve long-term continence.**

- **Bladder neck stenosis lowers continence**, mainly because **secondary procedures damage the external sphincter**.
- **Vesicourethral anastomotic stenosis (VUAS)** impairs coaptation. Its treatment can cause SUI because **the contracture extends into the membranous urethra**, which shows endoscopically as **loss of coaptation and radiating folds**.
- **Climacturia** (leakage with arousal or orgasm) is little studied; neurogenic mechanisms have been proposed.

Concept — Male continence runs on two sphincters in series. RP removes the internal one, which leaves the rhabdosphincter carrying continence alone. Every post-RP risk factor damages that sphincter, its innervation or its coaptation.

MECHANISM OF CONTINENCE WITH SURGICAL DEVICES

Compression Devices

- **AUS (AMS 800)**: a fluid-filled cuff around the bladder neck or bulbar urethra gives **circumferential compression over about 2 cm**. The **compliance of the pressure-regulating balloon (PRB)** sets the compression, and the PRB is chosen for tissue quality and cuff site.
- **Standard bulbar PRB is 61–70 cm H₂O**, the **balance between occlusion and erosion**. A lower pressure gives less continence and is used when erosion risk is excessive.
- **An open cuff removes the resistance**, so men **without detrusor contractility can still void by straining or Credé manoeuvre**.
- **ProACT**: paired silicone balloons beside the bladder neck give lateral compression. They are adjusted through two scrotal ports.

Male Slings

Nonadjustable Slings

- **Transobturator slings reposition and lengthen the membranous urethra without significant compression**.
- **Urodynamics: leak point pressure rises without obstruction**.
- **Fluoroscopy and magnetic resonance imaging (MRI): longer membranous urethra; elevated bladder neck, posterior bladder wall and rhabdosphincter**.
- **Virtue (quadratic fixation, four arms): compression against the urogenital diaphragm plus separate elevation** through prepubic and transobturator arms.

Adjustable Slings

- **ATOMS**: a cushion under the bulbar urethra compresses it, in men with a **functional sphincter**. It is topped up with saline through a scrotal port.

CONTINENCE DEVICES — MECHANISM AND ADJUSTABILITY

Device	Mechanism	Adjustment
AUS	Circumferential compression ; patient cycles cuff to void	None after implant (PRB fixed)
ProACT	Non-circumferential, lateral compression at bladder neck	Two scrotal ports
Advance / Advance XP	Repositioning, not compression	None
Virtue	Compression plus elevation	None

Device	Mechanism	Adjustment
ATOMS	Cushion compression of bulbar urethra	Scrotal port

- **What the table is for:** mechanism predicts candidacy. **Only the AUS does not depend on residual sphincter function.**
- **The trap:** the transobturator sling is **not compressive**. It works only if there is a functioning sphincter to reposition.

Concept — *Devices either replace the sphincter (AUS) or assist a residual sphincter (repositioning slings), and partial compressors sit between the two. This distinction drives every later selection decision.*

Figure legend.

- **Fig. 123.1 AMS 800:** cuff, pump and PRB. **The 3.5-cm cuff has a three-cushion design unique to that size.**

EVALUATION AND DIAGNOSIS

- **Initial evaluation:** history, examination, post-void residual (PVR), urinalysis. Cystoscopy and pressure-flow urodynamics (UDS) are added as indicated for VUAS, residual sphincter function and storage.

History

- Record **type, degree and severity**; prior surgery and **pelvic radiotherapy (RT)**; radiation cystitis; neurological symptoms.
- **Key task:** classify as SUI, UII, mixed or overflow. Leakage during sleep and nocturia help separate severe SUI from urgency.
- After CaP treatment: **disease status, planned adjuvant therapy, life expectancy.**

Physical Examination

- Examine the **abdomen, back, genitalia, perineum, rectum and nervous system.**
- **Assess manual dexterity before an AUS.** Treat skin breakdown or fungal/bacterial infection before surgery.
- **Previous incisions** guide PRB placement. **Inguinal hernia or prior repair** matters for an external-ring route. **Scrotal pathology** (hydrocele, hernia, mass) affects pump placement.

Patient Reported Outcome Measures

- **Patient reported outcome measures (PROMs)** quantify symptoms and bother and inform the choice of operation.
- **UCLA Prostate Cancer Index and Expanded Prostate Cancer Index Composite (EPIC)** capture post-RP SUI and UII. They are research tools, and their **clinical performance and responsiveness have not been assessed.**
- **Incontinence Symptom Index and Incontinence Impact Questionnaire-7** have been used for AUS outcomes but were **not developed for male SUI.**

Voiding Diaries

- **A 2–7-day diary** counts episodes, estimates **capacity, identifies urgency and UII** and helps separate SUI from UII.

Pad Weight Testing

- **Pad counts are unreliable** because the threshold for changing a pad varies. **The 24-h pad weight test is objective:** pre-weighed pads → worn 24 h → reweighed → **weight gain = leakage = severity.**

- **SUFU Pad Test** (Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction): a questionnaire on **number, size and wetness of pads over 24 h**. It **correlates with actual loss after RP** but is **not validated for predicting surgical outcome**.

FIG. 123.3 AUA/SUFU GUIDELINE EVALUATION ALGORITHM FOR INCONTINENCE AFTER PROSTATE TREATMENT (SANDHU ET AL., 2019)

Presentation	Pathway
Urgency or urgency-predominant mixed UI	American Urological Association (AUA)/SUFU overactive bladder guideline
SUI or stress-predominant mixed UI	Pelvic floor muscle exercises/therapy (PFME/PFMT) + goals → interval follow-up → improved, not bothered: no treatment; not improved or bothered: discuss surgical vs non-surgical → any choice may later lead to the surgical algorithm
Other pathology suspected (microhaematuria, low flow, high PVR)	Cystoscopy ± UDS → anastomotic stricture: treat, interval cystoscopy ; other pathology: treat → back to PFMT pathway

- **The logic:** exclude urgency-predominant disease and obstruction first. **Surgery follows only failed or unsatisfactory PFMT.**

Concept — *Evaluation proves the leak is sphincteric and measures how much. Type decides whether SUI surgery is relevant, and pad weight decides which device.*

Standing Cough Test

- A standing forceful cough (or Valsalva) in clinic demonstrates SUI and correlates with 24-h pad weight.

STANDING COUGH TEST — GRADE AND MEAN 24-H PAD WEIGHT (YI ET AL., 2020)

Grade	Leakage	Pad weight
0	Reported, not demonstrable	~57 g
1	Delayed drops	~117 g
2	Early drops , no stream	~223 g
3	Delayed stream	~385 g
4	Early, persistent stream	~513 g

- **What it is for:** a clinic proxy for pad weight. It is graded on **timing (delayed vs early)** and **form (drops vs stream)**. Grades 0–2 fall below 400 g; grades 3–4 lie in AUS territory.
- **The trap:** Grade 0 is not dry. The patient still loses about 57 g a day.

Laboratory

- Urinalysis and culture are mandatory. Investigate every abnormality and eradicate urinary tract infection (UTI).
- Creatinine (renal function) and PSA (cancer status). Evaluate renal insufficiency carefully before surgery.

Cystoscopy

- Mandatory when VUAS, bladder neck contracture or stricture is suspected, because **unrecognised obstruction compromises the outcome**.
- Assess residual sphincter function: poor coaptation → AUS, not a transobturator sling.
- Recurrent SUI after AUS: mechanical failure, fluid leak, atrophy.
- After explant for erosion, before any new AUS: exclude stricture and diverticulum.

Urodynamics

- **UDS measure capacity, compliance and contractility. Intrinsic sphincter deficiency is present in nearly all.**
- **Not routine. Indicated if the diagnosis is in doubt:** mixed UI, SUI not demonstrable, overflow, suspected poor compliance.
- **DO with mixed UI:** counsel that **UUI may persist after surgery.**
- **Poor compliance is the graver finding,** because **high-pressure storage damages the kidneys.** It does not reliably worsen AUS continence, so **upper-tract deterioration can proceed silently.**
- **Detrusor hypocontractility favours the AUS:** the cuff opens for voiding (straining or Credé), whereas a sling adds **fixed** resistance that a weak detrusor may not overcome.

KEY POINTS

- **History and examination** define type, severity and capacity in most men.
- **Uroflowmetry and PVR** screen for obstruction.
- **Obstruction, significant overactivity or poor contractility → UDS.**
- **Urethral, vesicourethral or bladder neck pathology → cystoscopy.**
- **Urinalysis (± culture) before surgery.**

Figure legend.

- **Fig. 123.4:** post-RP sphincter with a **quadrant defect at 9–12 o'clock and incomplete coaptation. Not an ideal transobturator sling candidate,** because the sling relies on repositioning a functional sphincter.

Concept — *Cystoscopy and UDS each answer a device-selection question. Cystoscopy asks whether there is a sphincter worth repositioning. UDS asks whether the bladder can empty against fixed resistance and store safely.*

INDICATIONS FOR SURGERY

- **Indication:** irreversible sphincter incompetence with UI affecting QoL.
- **Pelvic floor muscle training (PFMT) ± biofeedback or electrical stimulation works for post-RP SUI.** It shortens time to continence (better at 1 and 3 months with stimulation), with **no benefit beyond 12 months.**
- **Do not delay surgery for severe or gravitational SUI with no improvement after 6 months,** especially with a **cystoscopic sphincter defect.**
- **Use shared decision making,** weighing QoL gain against **performance status and life expectancy.**
- **AUS and perineal slings are first-line. Bulking agents are not in the post-RP or post-transurethral resection of the prostate (TURP) algorithm.** Reserve them for **small-volume leakage in men unfit for more.**
- **Selection factors:** severity, bother, **cognition, manual dexterity, body mass index (BMI), prior RT, prior surgery including urethroplasty, bladder function (e.g. neurogenic).**

CONTRAINDICATIONS TO MALE SUI SURGERY

Require an alternative to SUI surgery (absolute)	Relative
Low-compliance bladder threatening the kidneys (reflux at low pressure)	Frequent future transurethral management: bladder cancer, refractory VUAS, radiation cystitis
Low-capacity bladder (severe frequency or UUI)	Repeated instrumentation → erosion or infection

Require an alternative to SUI surgery (absolute)	Relative
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Tissue too poor for a sling or AUS → clamps, catheters, bladder neck closure, diversion	
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- **Radiation cystitis** often needs **large-bore catheters for clot irrigation**. Give **hyperbaric oxygen** and require **6 months free of refractory bleeding** before an AUS. Repeated clot retention → **cystectomy and diversion**.

Compression Devices

Artificial Urinary Sphincter

- **Standard for moderate-to-severe SUI with adequate cognition and dexterity**. It works **without detrusor contraction** and after RT, and has the **longest follow-up**.
- **Success predictors**: surgeon experience, high volume, intact compliance. **Multiple reoperations** are common.

ProACT

- **Fewer pads and better QoL**. In a randomised trial it matched slings for continence, **with more urethral erosion**. It does worse than slings after RT.

Slings

- **For mild-to-moderate SUI; non-adjustable or adjustable**. The commonest are the **Advance and Advance XP (transobturator)**. **Advance XP is durable**, whereas the **I-STP TOMS deteriorates progressively over 5 years**.
- **Adjustable slings have limited data**. **ATOMS beats ProACT** on dry rate, improvement, satisfaction and pad use.

DEVICE OUTCOMES — POOLED EFFICACY AND PRINCIPAL CAVEAT

Device	Pooled efficacy	Principal caveat
AUS (AMS 800)	Cure ~56%; social continence (0–1 pad/day) 55–77%; satisfaction ~90%	Revision 10–29% at 3 years ; needs dexterity and cognition
ProACT	Dry ~60%	More erosion than slings ; worse after RT
Transobturator sling	Cure ~60%	Fails with RT, severe UI, prior surgery
ATOMS	Dry ~67%; improved ~90%	Complications ~16% ; limited data

- **The pattern**: pooled dry rates cluster at 55–67% for every device. **What differs is durability, erosion and performance after RT**. Only the AUS has robust long-term data across all severities.
- **The trap**: ProACT's parity with slings hides **more erosion and poor post-RT results**.
- **Predictors of sling failure**: prior RT, greater severity, previous surgery.
- **Severity by 24-h pad weight**: mild <150 g, moderate <400 g. Transobturator slings are mainly for mild-to-moderate UI.
- **MASTER (landmark randomised controlled trial (RCT), AUS vs transobturator sling)**: the sling was non-inferior for continence at 12 months. The chapter stresses **AUS superiority with pad weight >250 g or prior RT**.
- **Slings remain an option in severe SUI** for men who **refuse an AUS** (fear of infection, erosion, failure) or have **limited physical or cognitive capacity**.

FIG. 123.5 AUA/SUFU ALGORITHM FOR SURGICAL MANAGEMENT OF INCONTINENCE AFTER PROSTATE TREATMENT (SANDHU ET AL., 2019)

Severity and RT	Guidance
Severe UI or any RT	AUS first-line; sling with appropriate counselling
Moderate, no RT	Discuss AUS, sling, possibly other devices
Mild, no RT	Discuss continued PFMT, sling, possibly AUS or other devices

- **Counselling points in the algorithm:**
- **AUS:** assess the physical and cognitive ability to operate it; **life span 8–10 years; one thin pad/day expected;** replaceable if it fails.
- **Sling:** **one thin pad/day; most failures go on to an AUS.**
- **Other:** **bulking rarely durable;** adjustable balloons only in **specialised centres.**
- **After surgery:** AUS failure → Fig. 123.12. Sling failure → Fig. 123.13. Other failure → repeat evaluation. Success → interval follow-up.

KEY POINTS

- **Operate for intrinsic sphincter deficiency failing conservative care.**
- **Low-compliance, low-capacity bladder or unusable tissue → alternative to SUI surgery.**
- **Relative contraindications:** bladder cancer, refractory VUAS, radiation cystitis.
- **Implant choice** weighs severity, prior surgery, bladder function, cystoscopy, dexterity, cognition, reoperation risk and preference.
- **Slings only very selectively after RT or in severe gravitational UI.**
- **No AUS with poor dexterity or cognition, poor urethral tissue or severely reduced compliance.**

Concept — Choice runs on two axes: severity and radiation. Whatever makes the residual sphincter less reliable (higher pad weight, RT, prior surgery, poor coaptation, weak detrusor) moves the choice from sling to AUS. Inability to work a pump moves it back.

SURGICAL DECISION MAKING IN THE HIGH-RISK PATIENT

- **"Fragile urethra"** = raised risk of AUS cuff erosion. Risk factors: **prior pelvic RT, previous failed AUS (erosion or infection), previous urethroplasty, hypogonadism.**

Prior Pelvic Radiotherapy

- **RT causes microvascular injury to the corpus spongiosum,** and the risk persists despite modern delivery.
- **Erosion occurs in 3.4–33% after RT vs 1–13% without it, and irradiated men erode sooner** (median ~1 vs ~3 years).
- **RT predicts the first explant but not later implant failures.** Those are driven by prior erosion and urethroplasty.

Previous Erosion/Infection

- **Coronary artery disease (CAD) and age >80 years increase erosion risk.**
- **Diabetes and smoking have not been shown to be independent risks,** but the series are small and may underestimate risk, so **optimise both.**
- **After one erosion, 5-year re-erosion approaches 50%.**

Previous Urethroplasty

- **An established, independent risk, and greater than RT** in a multi-institutional study.
- **Transection divides the bulbourethral arteries**, so the urethra loses its dual blood supply. **Non-transecting urethroplasty preserves them in theory; any benefit for erosion is unproven.**

Low Testosterone

- **An independent erosion risk. Urethral androgens probably support urethral health**, and most men who eroded in one series were hypogonadal.
- **Neither ADT (androgen deprivation therapy) as a risk nor testosterone supplementation as protection has been shown.**

Strategies to reduce erosion in high-risk men.

- **Optimise medically** for wound healing and urethral health.
- **Commonest erosion site is disputed:** a single-institution series found mainly ventral or lateral erosions, a multi-institutional series roughly equal dorsal, ventral and lateral.
- **Transcorporal (TC) AUS (Webster, 2002)** puts tunica albuginea between the dorsal urethra and the cuff. In theory it preserves vascularity and **adds bulk where the spongiosum is thinnest. Evidence is mixed.**
- **Reduced-pressure PRB (51–60 cm H₂O) or a cuff 0.5 cm larger than measured:** supporting data are scarce.
- **Many men still prefer repeated AUS attempts** to pads, a catheter or diversion.

KEY POINTS

- **Erosion risks:** RT, failed AUS, urethroplasty, hypogonadism.
- **Optimise medically; consider TC cuff and reduced-pressure PRB.**

Concept — *All high-risk factors converge on one pathway: an ischaemic spongiosum under a compressive cuff. Each countermeasure adds tissue (TC cuff, larger cuff), lowers pressure (51–60 PRB) or preserves supply (optimisation, non-transecting urethroplasty).*

TECHNIQUE OF IMPLANTATION OF CURRENTLY APPROVED AND AVAILABLE DEVICES

Operative Preparation for All Implant Surgeries

- **Sequence:** general or spinal anaesthesia, low lithotomy → shave → chlorhexidine–alcohol prep of lower abdomen, genitals and perineum → intravenous antibiotics within 60 min: aminoglycoside 5 mg/kg + first- or second-generation cephalosporin or vancomycin (AUA Best Practice Statement) → drape → ± urethral catheter.
- **A 10-min chlorhexidine scrub then iodine povacrylex–alcohol (DuraPrep)** gives fewer positive skin cultures than povidone–iodine, **but no fewer implant infections.**

Artificial Urinary Sphincter

Bulbar Cuff Artificial Urinary Sphincter

- **Perineal incision; cuff as proximal as possible on the bulbar urethra, proximal to corporal fusion.** This allows safe circumferential dissection, protects the cuff from opening when sitting and gives the **widest spongiosum.**
- **Sequence:** skin → Colles fascia → bulbospongiosus → incise Buck fascia where it reflects onto the diverging corpora → 2-cm tunnel (cuff width) dorsal to Buck fascia over the urethral roof, sharp and

under vision → right-angle clamp → **measure circumference (usually 4.0–4.5 cm)** → place cuff → tubing through bulbospongiosus into the **deep perineal space beneath Colles**.

- **Sizing: consistency of method matters more than catheter in or out.** The cuff fits slightly tighter than the tape; if obviously wrong, move to the next size. **Looser cuffs give better continence. Best test: visual and endoscopic appearance.**
- **Sizes run 3.5–11 cm** (0.5-cm steps to 8.0, then 9, 10, 11). **The large sizes are for the bladder neck.**
- **3.5-cm cuff:** for atrophy after a previous AUS or very distal placement after revisions. It erodes more after RT, so prefer TC and/or a 51–60 cm H₂O PRB.

Pressure-Regulating Balloon

- **Scrotal, perineal or abdominal route**, chosen by previous incisions, habitus and preference.
- **Abdominal:** lower-quadrant incision **ipsilateral to the pump** → incise rectus or external oblique fascia → split muscle → **preperitoneal (or intraperitoneal) space** → digital pocket → close fascia around the tubing.
- **External inguinal ring (penile-implant technique)**, from a subinguinal or perineal incision: through the floor of the inguinal canal into the retropubic space. **Ectopic submuscular: above transversalis fascia, behind the posterior rectus sheath, no sutures** (transversalis fascia prevents extrusion).
- **Avoid the ring and ectopic routes after mesh hernia repair, radical cystectomy or extensive abdominal surgery. Use the abdominal approach**, which lowers bladder and bowel injury.

PRB PRESSURE BY INDICATION

PRB (cm H ₂ O)	Use
51–60	Fragile urethra (RT, atrophy, prior erosion)
61–70	Standard bulbar cuff
71–80	Bladder neck cuff

- **Pressure rises with the robustness of the tissue under the cuff. Trap:** raising PRB pressure to treat atrophy increases erosion.
- **Fill with 23 mL; bladder neck cuffs need more** (per manufacturer). **Solutions:** 0.9% saline; Cysto-Conray 60 mL + 15 mL sterile water; Hypaque 25% 50 mL + 60 mL sterile water. **Always dilute contrast with sterile water (isotonic). No contrast with iodine or contrast allergy.**

Control Pump Placement

- **Anterior scrotum**, through the inguinal, scrotal or perineal incision.
- **With an abdominal PRB:** dissect down **above the abdominal fascia, deep to Scarpa** → a finger invaginates the **dependent scrotal skin into the incision** → strip fascia off dartos → **subdartos pocket**.
- **Single-incision:** subdartos pouch directly, pump oriented **towards the foot (scrotal) or towards the head (perineal)**.

Making the Connections

- **Connect in the abdominal incision when there is one**, which protects the connections and eases later exploration. **Quick connectors** are secure and watertight.
- **Cycle the device several times, then urethroscopy:** closed cuff = slight blanching (adequate coaptation, filling, connection); deactivated = minimal coaptation; excessive coaptation → larger cuff.

Concept — *First-time technique follows one rule: go proximal. That site gives the thickest spongiosum and a cuff protected when sitting, and it saves the distal urethra for the revisions most men will need. Sizing and PRB choice then tune compression against ischaemia.*

Tandem Cuff Artificial Urinary Sphincter

- **Indication:** persistent significant UI after a single cuff, from incomplete occlusion (a minority of men) → second bulbar cuff.
- **Sequence:** via the perineal incision → circumferential dissection ≥ 1.5 –2 cm distal to the primary cuff → new cuff → divide the existing tubing and join with a metal Y-connector → add 3 mL → device deactivated for 6 weeks.

Transcorporal Cuff

- **Indications:** difficult primary or revision surgery; urethra <3.5 cm. After erosion or urethroplasty, use a different site along the bulbar urethra.
- **Sequence:** parallel longitudinal incisions in the corporal tunica albuginea near the spongiosum → dissect through the corporal septum from side to side → ventral tunica stays on the dorsal spongiosum → imbricate the lateral tunical edges behind the urethra (less haematoma) → measure.
- **Erectile function:** the risk is theoretical; function is preserved in most sexually active men in small series.
- **High-risk series:** most cured or improved, no erosions. Overall evidence is mixed because most series combine techniques. Especially useful when proximal atrophy leaves only a small distal urethra.
- **Variants:** "Gullwing", a complete tunical wrap through wider incisions, gives circumferential rather than dorsal-only protection in theory but has no long-term data. A cuff around the bulbospongiosus protects the ventral urethra: high 5-year revision but erosion very rare.

Transscrotal Artificial Urinary Sphincter

- **Wilson (2003):** all components through one transverse scrotal incision. It was designed for reoperation after atrophy, allowing a distal tandem or TC cuff.
- **Continence is inferior**, because the approach cannot reach the proximal, wider bulbar urethra: perineal cuffs are mostly 4.5 cm, transscrotal mostly 4.0 cm. The modified enhanced-exposure version (2010) allows larger proximal cuffs.
- **The authors prefer the perineal route, which keeps distal sites for revision.**

Bladder Neck Artificial Urinary Sphincter

- **Optional, invasive and harder**, for sphincteric UI with an intact prostate (e.g. myelomeningocele, neuropathic disorders). Contraindicated after RP.
- **Open:** lower abdominal incision → expose bladder base → circumferential dissection of the vesicoprostatic junction. **Robotic:** dissects the bladder–seminal vesicle plane under vision and avoids that incision.
- **Adults:** cuff usually ≥ 8 cm, PRB 71–80 cm H₂O, more fluid. Children may need very different specifications.

AUS CUFF APPROACHES COMPARED

Approach	Advantage	Drawback
Bulbar perineal (standard)	Proximal, thickest spongiosum; saves distal sites	—
Tandem	More occlusion	More distal-cuff erosion; uses revision sites; not first-line
Transcorporal	Less urethral injury at reoperation (prior erosion, injury, urethroplasty, rectourethral fistula repair); better fit (bulk); possible erosion reduction (unproven)	More transient retention needing suprapubic cystostomy
Transscrotal	Single incision	Inferior continence (smaller, more distal cuff)

Approach	Advantage	Drawback
Bladder neck	Less erosion and atrophy	Invasive; contraindicated after RP

- **The pattern:** every alternative trades the proximal, well-vascularised site for **bulk or extra occlusion**. **Trap:** the bladder neck cuff erodes least but is barred after prostatectomy, the commonest indication.

KEY POINTS

- **Bulbar cuff proximal to corporal convergence.**
- **Bulbar PRB 61–70 cm H₂O;** abdominal, subinguinal or perineal route.
- **Deactivate 4–6 weeks after surgery** so the urethra heals without erosion.
- **Non-mechanical revision:** TC, 3.5-cm or tandem cuff.

Concept — *Each alternative cuff strategy spends urethra or adds risk to buy occlusion. First-time placement therefore stays single, proximal and perineal, with the others held in reserve.*

MALE SLINGS

Transobturator Bulbourethral Sling

- **First described by Rehder and Gozzi (2007), to reposition and lengthen the membranous urethra.**
Indications: SUI after CaP treatment, TURP or laser enucleation. **Advance XP** (improved anchoring) has replaced the Advance.
- **Sequence:** 1. Perineal incision → **open bulbospongiosus in the midline** and dissect it off the spongiosum to the **perineal body** → **mark this level** → **release the proximal bulb by 1–2 cm** so it can be repositioned. 2. Develop the space between spongiosum and diverging corpora. **Feel the inferior pubic ramus behind each crus** so the trocar lands on the fingertip. 3. **Stab incision ~1 fingerbreadth below adductor longus, in the groin crease** → spinal needle locates the **obturator foramen just lateral to the pubic ramus**. 4. **Helical passer, outside-in:** thigh → foramen → perineal wound medial to the corpus. **Two "pops"**, then turn. **Exit high in the angle between spongiosum (medial) and inferior pubic ramus (lateral)**. 5. Mesh positioned loosely and pulled to the thigh **without tension**. **Proximal edge at the marked distal perineal body**. 6. **Fix to the spongiosum with four absorbable sutures**. 7. **Pull both arms equally under cystoscopy until the bulbar mucosa coapts**. The **perineal body and bulb move 3–4 cm proximally and cephalad**. 8. Remove sleeves → trim at skin, bury or tunnel → **close bulbospongiosus, Colles fascia, skin**.

Quadratic Sling

- **Virtue:** two transobturator strips elevate, two prepubic arms compress. **Prepubic fixation is important.** A small **mild-UI** cohort was **durable at 3 years**. Its role is undefined, but because it **compresses** it **may suit wider severities and aetiologies**.

Adjustable Continence Devices

- **ProACT sequence:** catheter; **contrast fills the bladder** to show the neck on fluoroscopy → **sharp trocar** through a perineal incision → **exchange for a blunt trocar beyond the pelvic diaphragm** → advance to the bladder neck → **U-channel sheath deploys the balloon**, filled with saline → repeat on the other side → **titanium port in a scrotal subcutaneous tunnel** → close; remove catheter.
- **ATOMS sequence** (dissection as for Advance XP, but **bulbospongiosus exposed, not opened**): palpate crura, find the foramen → **needles through the superomedial corner of the foramen**, around the ischiopubic ramus, **out through the triangle of bulbospongiosus (medial), ischiopubic ramus (lateral)**

and transverse perineal muscle (posterior) → suture through the needle eye, withdraw; repeat on the other side → tie the arms to the sutures and pull through → **cushion centred under the bulbar urethra** → fix mesh arms to the cushion (supplied polypropylene sutures), trim → **subdartos scrotal port filled with 9 mL of saline** → close.

KEY POINTS

- **Slings:** mild-to-moderate SUI, intact contractility, healthy non-irradiated urethra.
- **Transobturator slings** need an intact external sphincter.
- **Quadratic and adjustable slings:** limited efficacy and durability data.

Concept — The transobturator sling works by relocation, not squeeze, so its technique centres on freeing the proximal bulb and tensioning to endoscopic coaptation. Only the compressive designs (Virtue, ATOMS, ProACT) might reach beyond men with an intact sphincter.

EVALUATION OF PERSISTENT INCONTINENCE AFTER ARTIFICIAL URINARY SPHINCTER AND SLING

Incontinence After Previous Artificial Urinary Sphincter

- **Differential:** inadvertent deactivation; **insufficient compression** (oversized cuff, urethral or bladder neck atrophy); **mechanical failure with fluid loss; erosion; storage failure**. Rare causes: **plugged delay-fill resistor, kinked tubing**. The work-up follows Fig. 123.3.
- **Onset is the first clue:** sudden → deactivation or mechanical failure; slow progression → atrophy.

RECURRENT INCONTINENCE AFTER AUS — DIFFERENTIAL DIAGNOSIS

Cause	Clue	Test / action
Inadvertent deactivation	Sudden	Active cycling excludes it. Reactivate: squeeze pump's lateral edges , cotton-tip pressure opposite the button, or hold the button ≥30 s
Mechanical failure, fluid loss	Sudden	PRB imaging during cycling: plain film (contrast) or ultrasound (saline)
Atrophy or oversizing	Gradual	PRB changes size and refills passively → atrophy, not failure; cystoscopy of cuff during cycling
Erosion	Variable	Cystoscopy
Storage failure or DO	Urgency	UDS

- **Isolated PRB replacement** has been proposed for an intact system without atrophy (**intrinsic PRB failure**). **Exclude PRB herniation**, because the PRB needs a **deep pelvic or submuscular position**.

FIG. 123.12 AUA/SUFU ALGORITHM FOR AUS FAILURE (SANDHU ET AL., 2019)

Presentation	Management
Infection	Explant urgently; replacement possible at 3–6 months
Erosion	Explant urgently; catheter ~3 weeks; replacement possible at 3–6 months
Worsening UI, no infection or erosion	Imaging of PRB volume → little or no fluid: replace the entire AUS or failed parts; normal (>20 mL): downsize, move or add a cuff, or raise PRB pressure

- **The logic:** infection or erosion → urgent explant. Otherwise **PRB volume separates device failure from tissue failure**. **Trap:** the guideline lists raising PRB pressure, but the text warns that **it increases erosion**.

Incontinence After Previous Sling

- **Work-up as for AUS failure, plus sling type (Advance, Virtue, bone-anchored) and timing.**
- **Advance, early loosening** (improved, then recurred **within 6 weeks**) → **re-explore**: cut each arm lateral to the bulbar urethra → **remove the central portion from the spongiosum** (extensive dissection if it was elevated and invaginated) → new sling.
- **Persistent or delayed SUI: separate DO from SUI. Sling failure → AUS (gold standard), with outcomes equal to a first-time AUS. Cystoscopy for all with worsening SUI (mesh erosion). A second Advance for late failure (slippage) succeeds in about half.** The authors reserve it for men in whom an **AUS is contraindicated or refused**.

FIG. 123.13 AUA/SUFU ALGORITHM FOR SLING FAILURE (SANDHU ET AL., 2019)

Presentation	Management
Infection or erosion	Explant as much as possible; consider AUS at 3–6 months
Inadequate continence	AUS ($\pm$ sling explant), or sling replacement by trained providers only

KEY POINTS

- **AUS failure: separate deactivation, under-compression, mechanical failure, erosion, storage failure.**
- **Cystoscopy excludes erosion and shows the cuff cycling (atrophy).**

Concept — *The clue is tempo. Sudden failure means the device has failed; gradual failure means the tissue has. PRB fluid volume separates the two, and infection or erosion overrides both.*

ARTIFICIAL URINARY SPHINCTER COMPLICATIONS

Urinary Retention

- **Rare; commoner with TC and 3.5-cm cuffs.**
- **Sequence: confirm deactivation → small catheter for 24–48 h → if the second voiding trial at 48 h fails → suprapubic cystostomy** (less erosion risk), **placed under ultrasound or fluoroscopy** so the PRB is not punctured.
- **Retention persisting for weeks = undersized cuff** → replace it, correlating with preoperative UDS. **Late-onset retention → endoscopy and UDS** for proximal obstruction, erosion or detrusor failure.

Artificial Urinary Sphincter Infection

- **1–3% after a first AUS; up to 10% in high-risk men. Organisms: skin flora, *Staphylococcus epidermidis* and *S. aureus*.**
- **InhibiZone (rifampicin + minocycline) has not reduced AUS infection** as it has for penile implants.
- **Late infection (>4 months): indolent organisms from surgery or haematogenous spread → antibiotic prophylaxis before urinary tract manipulation (AUA).**
- **Scrotal pain is the key early sign ($\pm$ erythema, oedema, pus). Implant infection does not respond to antibiotics, so explant.**

AUS INFECTION — DELAYED VERSUS IMMEDIATE SALVAGE

Traditional	Immediate salvage (infected, non-eroded)
Remove device → wait several months → reimplant	Remove all → seven-solution antiseptic irrigation (penile prosthesis protocol) → immediate new device; most salvaged

Traditional	Immediate salvage (infected, non-eroded)
Default	Contraindicated with sepsis, ketoacidosis, necrotising infection, erosion, immunosuppression, gross pus

Urethral Erosion

- **2–15% of implants; up to 35% with multiple risk factors.** Delayed activation has lowered the rate, especially after reimplantation. **Catheterisation (± deactivation) is itself a risk.**

EROSION RISK FACTORS — THE CHAPTER'S TWO SECTIONS RECONCILED

Factor	High-risk section	Erosion section
Prior RT, prior erosion, urethroplasty	Strong	Strong
CAD	Risk	Risk
Age >80 years	Risk	—
Smoking, diabetes	Not independently shown	"Well established"
Obesity, hypertension	—	Listed
Hypogonadism	Shown	Mixed
Prior revisions	—	Mixed
Catheterisation	—	Risk

- **Why they differ:** the high-risk section counts **independent (multivariable) risk**; the erosion section lists **comorbidity associations**. **Optimise all of them.**
- **Explant every component, as all are assumed infected** (a retained PRB becomes infected in only a minority). **Treat the perineal wound as infected:** loose closure, or **secondary intention if grossly contaminated.**
- **Erosion site varies** (see the High-Risk section). **Stricture follows erosion in 8–35%.**

MANAGEMENT OF URETHRAL EROSION — OPTIONS AND STRICTURE OUTCOME

Strategy	Stricture outcome
Catheter only → urethrogram before removal	Traditional; high stricture rate when urethral loss is extensive
Pseudocapsule closure	High in one series; only prior RT (not erosion size) predicted stricture
In situ repair without mobilisation	Fewer strictures than catheter alone
Mobilisation + spatulated anastomotic urethroplasty	Fewer strictures than catheter alone with >50% loss

- **A multi-institutional counterpoint:** complete erosion strictures more often, and catheter, urethrorrhaphy (capsule repair) and anastomotic urethroplasty did not differ significantly.
- **Rule:** <50% urethral loss → catheter; >50% → unresolved. Balance stricture risk against contamination and the harm of mobilising in an infected field.
- **Reimplantation:**
 - Wait **3–6 months** and confirm patency (cystoscopy or retrograde urethrogram).
 - **New cuff proximal or distal to the erosion site, never at it** (scar, poor blood supply).
 - **Early re-erosion reaches 35%**, mostly within about 7 months.
 - **Nightly deactivation after reimplantation for prior erosion** gives high success with low recurrence.
 - **TC and/or a low-pressure PRB** reduce injury at reoperation and may protect against re-erosion.

Urethral Atrophy

- **Chronic compression of the spongy tissue under the cuff.** Onset is **insidious and slow** (unlike device failure). It is **one of the commonest reasons for revision**.
- **Options: downsize; move proximal or distal to thicker spongiosum; TC; tandem. A higher-pressure PRB alone increases erosion.**
- **Authors: downsize at the same site if possible. If the cuff is already 4.0 cm, move it proximally, or distally with TC, and measure afresh.** In a large revision series **most cuffs were downsized and distal relocation was the commonest move.** Downsizing improves pad use markedly.
- **Subcuff atrophy is contested.** One high-volume centre **treats every failure as mechanical and replaces the cuff at the same size**, with acceptable results. **Pressure profilometry confirms true atrophy in only about one-fifth of failures.**

Mechanical Failure

AUS DEVICE LIFE — BODY TEXT VERSUS GUIDELINE FIGURE

Source	Device life	Reason
Chapter text	7–10 years	Authors' synthesis of device-survival data
AUA/SUFU (Fig. 123.5)	8–10 years	Guideline counselling figure

- **No infection or erosion, and within 3 years of implant → replace the failed component alone.** Early failures are **mostly pump or cuff**, at a median of **≤1 year**.
- **A slow PRB leak is hard to confirm in theatre, so if in doubt replace everything. After 3 years most failures are the PRB, so replace the whole device.** An ohmmeter or burette test of the PRB may let experienced implanters avoid this.

KEY POINTS

- **Failed second voiding trial at 48 h → suprapubic cystostomy.**
- **Match erosion management to severity and contamination.**
- **After erosion, TC and/or low-pressure PRB.**
- **Replace the entire AUS if >3 years old (PRB failure).**

Concept — *Nearly every AUS complication arises at the cuff–spongiosum interface. Too tight or too long compressed gives atrophy, then erosion; a breach gives infection. Management therefore protects the urethra before the device: remove everything, wait, move the site, lower the pressure.*

SPECIAL CIRCUMSTANCES

Managing Device Implantation With Spongiosal Vascular Insufficiency

- **A fragile urethra erodes more readily, so protect its blood supply. Stricture needing repair before an AUS → non-transecting urethroplasty** (preserves the bulbourethral arteries). **Use TC AUS or a lower-pressure PRB.** Testosterone replacement in hypogonadal men is investigational.

Urethral and Vesicourethral Anastomotic Stricture

- **A stricture at the cuff site reflects compression or ischaemia and may herald erosion.** Periodic dilation has managed one such stricture.
- **Stricture proximal to an AUS: balloon dilation under vision or holmium laser through a flexible cystoscope or ureteroscope.**

- **ROBUST III (landmark RCT):** the US Food and Drug Administration (FDA)-approved paclitaxel-coated balloon vs direct vision internal urethrotomy (DVIU)/dilation. Freedom from intervention at 1 year: 83.2% vs 21.7%. Functional success 67% at 3 years (recurrent bulbar strictures). No VUAS data.
- **Post-erosion stricture repaired by urethroplasty:** nearly all men got another AUS, mostly TC, and about a quarter re-eroded by 2 years.
- **VUAS after AUS:** balloon or laser urethrotomy via ureteroscope. A 22–24 Fr scope proximal to the cuff → uncouple the cuff surgically first. Refractory → robotic VUAS reconstruction, which spares the cuff (AUA guideline).

VUAS AND AUS — SEQUENTIAL VERSUS SYNCHRONOUS TREATMENT

Traditional	Newer
Correct VUAS before AUS, avoiding treatment of recurrence with a cuff in place	Non-obiterated VUAS (<12 Fr) dilated at the same sitting as AUS
—	No difference in later VUAS treatments or in erosion

Urinary Incontinence After Cystectomy and Neobladder Reconstruction

- SUI after orthotopic neobladder: 33–95%, and QoL depends on regaining continence.
- An AUS improves most men, but most need revision (over a third of revisions for erosion or infection). Contributors: prior RT, bacterial colonisation, clean intermittent catheterisation (CIC).
- PRB ectopic via a lateral counter-incision, behind the rectus, to avoid neobladder and bowel.

Dual Implantation of Artificial Urinary Sphincter and Inflatable Penile Prosthesis

- An AUS and an inflatable penile prosthesis (IPP) have similar device survival and revision whether placed synchronously or metachronously. On meta-analysis, reoperation is ~25% either way, with high satisfaction. Counsel that dual implants need more revisions than a single IPP or AUS.

DUAL AUS + IPP — SYNCHRONOUS VERSUS METACHRONOUS

Synchronous	Metachronous
Penoscrotal (IPP) + perineal (proximal bulbar AUS) incisions	AUS first; IPP after activation
Safe and effective	Patients prefer continence before ED treatment

- **IPP already present:** standard bulbar AUS. If TC is needed, dissect between the tunica and the prosthesis pseudocapsule, and close the corporotomy with allograft (e.g. Tutoplast) for haemostasis.
- **Dual implants after neobladder:** no difference from age-matched controls in infection, mechanical failure or revision.

Climacturia

- **Orgasm-associated UI** in 20–68% after RP, including leakage on arousal.
- **Condoms and precoital voiding** help little. The authors use a sling or AUS according to the degree of SUI and climacturia.
- **Mini-jupette sling** (at IPP placement for ED): 3-cm corporotomies → graft (e.g. Tutoplast) sutured to the medial edges → inflation tightens the graft across the urethra, compressing it during sex. With mild SUI, leave the IPP partly inflated. Single-institution 5-year data show durable improvement in most, and no erosion or infection.

KEY POINTS

- **VUAS after AUS:** balloon or ureteroscopic laser.

- **Synchronous AUS + IPP is safe.**
- **Mini-jupette at IPP:** ED with climacturia and mild SUI.

Concept — *One rule unites these cases: protect the spongiosal blood supply and keep the cuff out of the path of instrumentation. Procedures are sequenced (continence first, strictures treated first or the cuff uncoupled) so the cuff never meets an unplanned large-bore scope or catheter.*

SLING COMPLICATIONS

Nonadjustable Slings

- Pain and wound problems are the commonest complications (meta-analysis). Inguinal pain from trocar passage is self-limiting: median visual analogue scale (VAS) score 0 at 3 months.

Adjustable Slings

- ATOMS complications are commoner, which may limit adoption.

Noncircumferential Compression Devices

- ProACT: bladder or urethral perforation is the commonest complication.

COMPLICATIONS OF SLINGS AND NON-CIRCUMFERENTIAL COMPRESSION DEVICES

	Non-adjustable sling	ATOMS	ProACT
Chronic pain	1.3%	1.5%	—
Erosion	0–0.4%	10% (with infection)	3.8%
Explant / revision	0–0.4%	10–15.8%	22%
Other	Urgency or retention 4.3–10.3%	—	Perforation 5.3%; leak 4.1%; migration 6.5%

- **The pattern: complications rise with adjustable hardware**, and the non-adjustable mesh sling is safest. **Trap:** "adjustable" is not gentler; **ATOMS and ProACT erode and need revision far more often.**

GAPS AND FUTURE DIRECTIONS

Innovations and Emerging Concepts in Devices

- The AUS works well but often needs multiple revisions, with significant morbidity. Slings and compression devices often succeed but have frequent complications and often miss patients' goals.
- **Future:** electronic control, patient-specific dynamic pressure control, new devices, to improve durability and widen eligibility.

Registries

- **Most long-term data are small, retrospective and single-institution. Registries and multi-institutional prospective studies are needed.** Trials are recruiting: AUS vs sling, ATOMS vs AUS, Virtue.

Surgical Training

- **More than 150,000 men have had an AUS or sling, yet many urologists never perform SUI surgery.** Fellowships and simulation should widen access.

SUMMARY

- **SUI surgery stays important** because localised CaP is still often treated surgically.
- **Assessment: history, examination, pad weight, uroflowmetry, PVR; cystoscopy and UDS selectively.**
- **The AUS is the most established treatment for moderate-to-severe SUI**, especially in complex cases.
- **Slings suit mild-to-moderate UI without prior urethral surgery, RT or detrusor acontractility.** Results after RT are poorer, and **neurogenic or post-traumatic use needs evidence.**
- **Comparison is confounded** by baseline severity, endpoints, follow-up and selection bias. **MASTER is the one RCT comparing them directly.**
- **Almost 90% of men report better urinary QoL after surgery, whatever the severity.**
- **Erosion and mechanical failure drive the search for alternatives.**